Evaluating Rating Reliability: A Cross-Platform Analysis of Movie Ratings and Review Sentiment

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Introduction

Background

Ratings differ across platforms influenced by users, systems, and environments
Are these ratings reliable, and do they reflect audience perception consistently?

Research Focus

Compare consistency of ratings across platforms
Examine alignment between ratings and review sentiment
Analyze patterns of user rating behavior

Approach

1. Compare TMDb and IMDb ratings
2. Analyze user behavior with MovieLens
3. Apply sentiment analysis on TMDb reviews

This project provides a practical perspective on how platform ratings reflect user perception and how they should be interpreted in decision-making.

Data sources

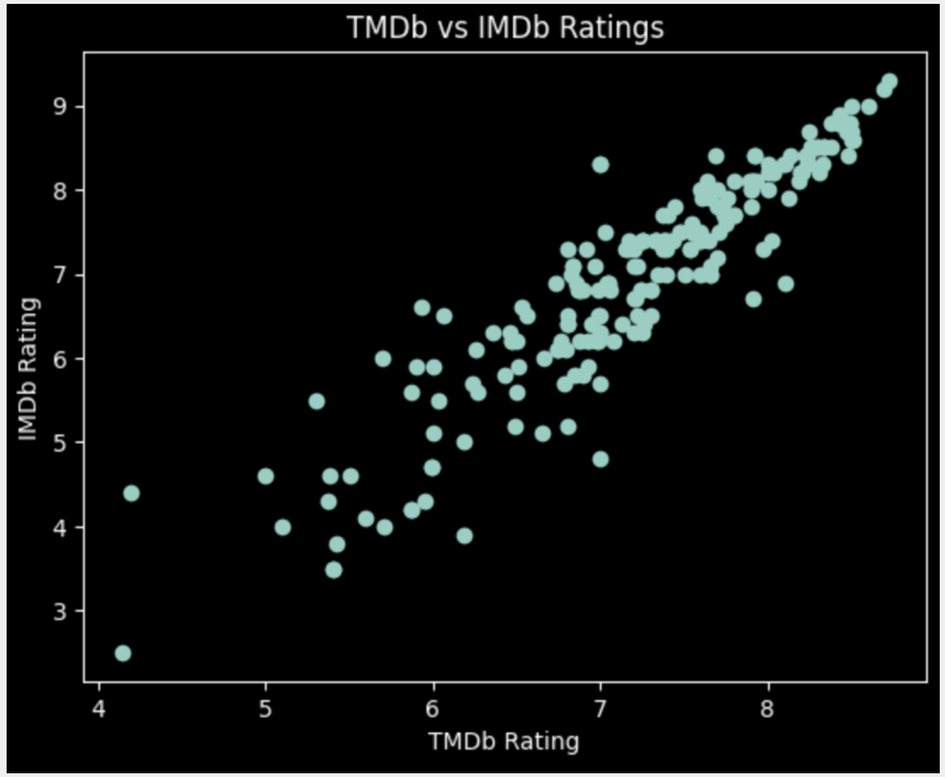
Source	Type	Source URL	List of Fields	Format	Purpose	Final Scale
TMDb	API	https://www.themoviedb.org/	ID, Title, Vote Average, Vote Count, Release Date, Genres, Review Text	json	Platform ratings + sentiment	300 movies collected; ~580 reviews
IMDb	Dataset	https://developer.imdb.com/non-commercial-datasets/	Movie ID, Title, Rating, Votes, Release Year	tsv	Platform comparison	48,189 cleaned records
MovieLens	Dataset	https://grouplens.org/datasets/movielens/1m/	Movie ID, Title, Rating, Timestamp	dat	User behavior analysis	3,706 cleaned movies

Source	Type	Data Used	Purpose	Final Scale
Merged TMDb + IMDb	Processed dataset	Matched ratings	Rating consistency	196 movies
Sentiment dataset		TMDb reviews	Sentiment-rating alignment	72 movies after filtering; ~580 reviews

Summary of the Results

1. Consistency of Ratings Across Platforms
(TMDb vs IMDb)

TMDb vs IMDb Ratings



Matched movies: n = 196

- Correlation = 0.92

suggesting that the two platforms usually move in the same direction.

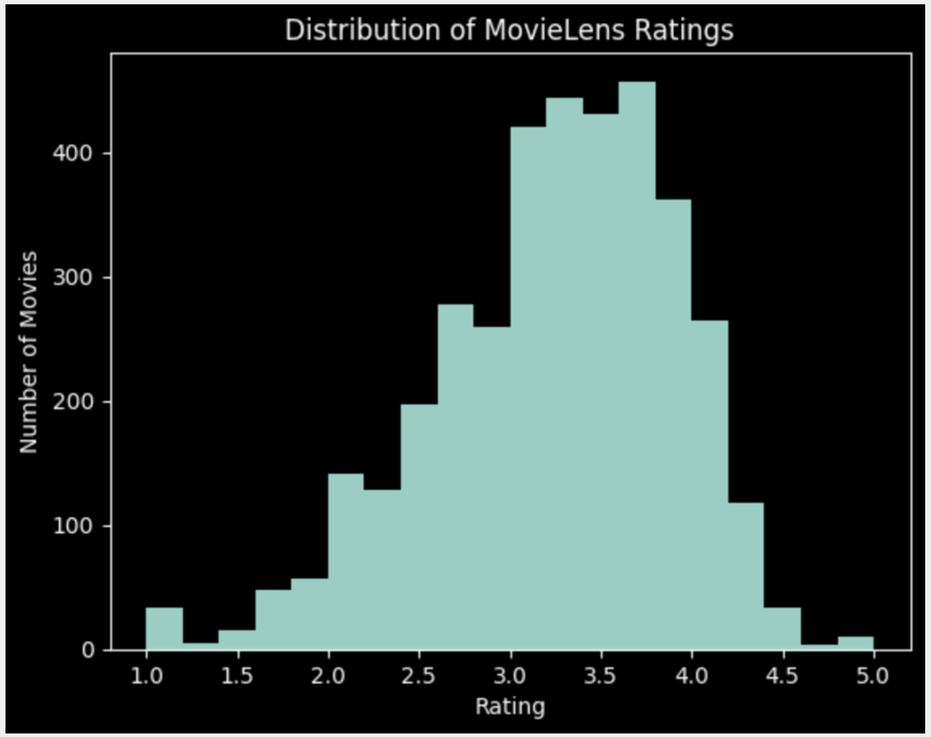
- Strong positive relationship between TMDb and IMDb ratings
- Higher consistency for highly rated movies
- Greater variation for lower-rated movies
- Overall differences are small

Summary of the Results

2. User rating behavior and sentiment tendency

(MovieLens + TMDb sentiment)

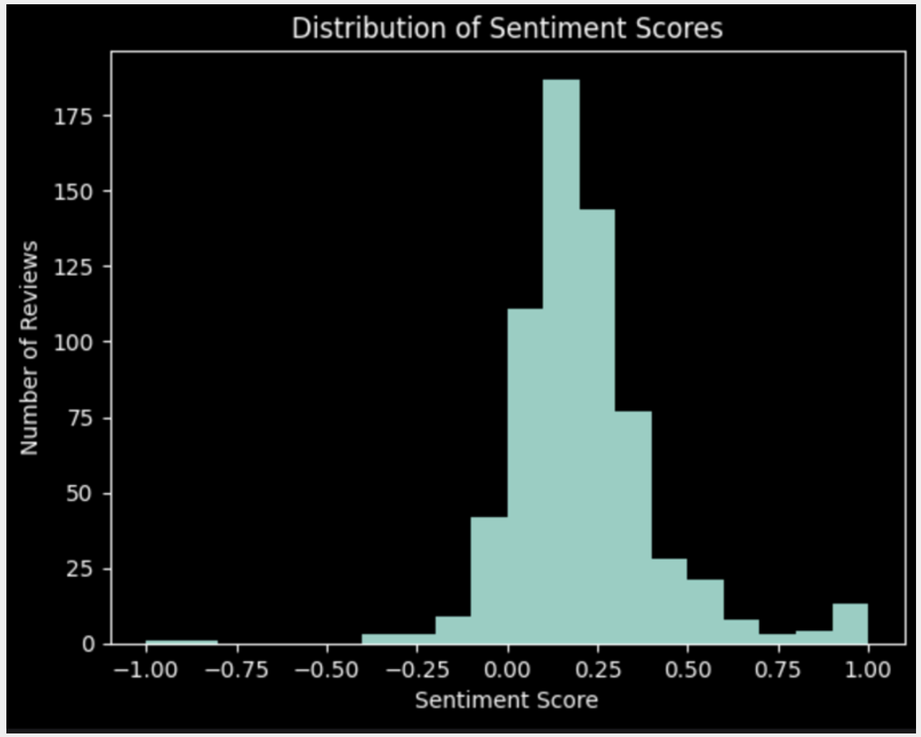
MovieLens rating distribution



MovieLens cleaned movies: n = 3,706

- Ratings concentrated between 3 and 4
- Users tend to give moderate scores
- Popularity does not necessarily lead to higher ratings

Sentiment distribution



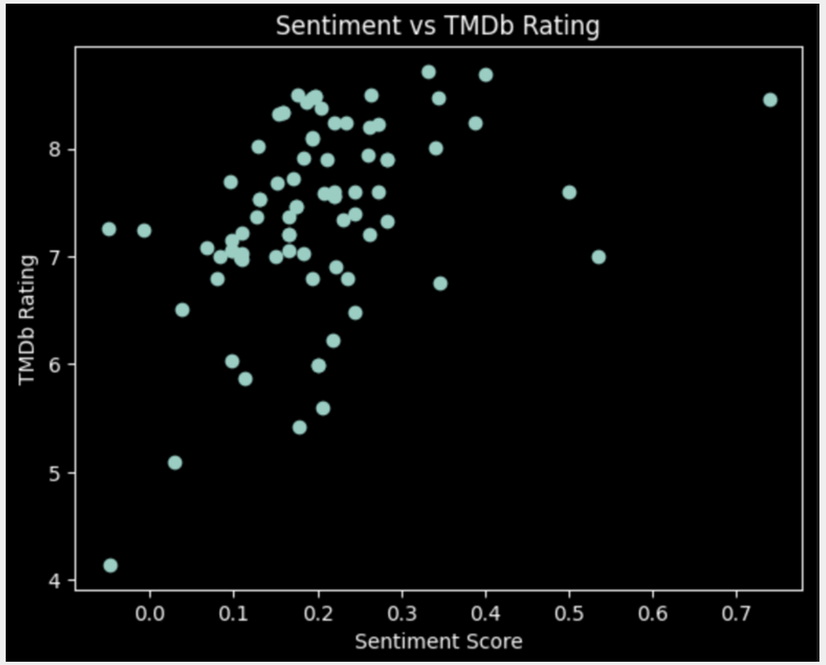
TMDb reviews: n ≈ 580

- Most sentiment scores are positive (>0)
- Reviews are generally favorable but not extreme

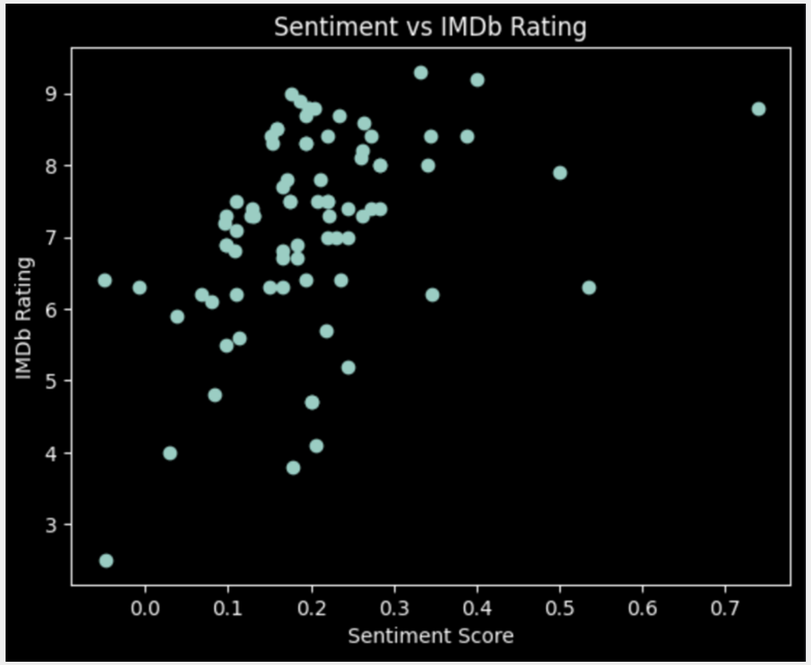
Summary of the Results

3. Alignment Between Sentiment and Platform Ratings

Sentiment vs TMDb



Sentiment vs IMDb



sentiment-matched movies n = 72

- This suggests that sentiment provides directional support for ratings, but it cannot fully explain rating differences.

- General positive trend observed
- Higher sentiment tends to correspond to higher ratings

Limitations

- General positive trend, but weak linear relationship
- Sentiment values cluster in a narrow range
- Review availability varies by movie

Challenges and Key Takeaway

- **Challenge 1: Data accessibility**

Some platforms had limited or unstable access to review data, so the project had to rely on more stable and structured sources.
This affected the final research design and made TMDb the main source for review sentiment.

- **Challenge 2: Sentiment interpretation**

TextBlob provides a simple lexicon-based sentiment score, which is useful for identifying general positive or negative tendencies.
However, it may not fully capture sarcasm, complex opinions, or detailed emotional nuance in long movie reviews.

- **Challenge 3: Cross-platform matching**

Matching movies across TMDb and IMDb required using title and release year.
This process can be affected by title formatting, remakes, alternate titles, and different release dates across platforms.

- **Key takeaway**

Platform ratings are useful indicators, but they should be interpreted with caution because data access, platform differences, and sentiment method limitations can influence the results.